

A sub-Mixon Grassmannian Frame Packing for Cell (d=3, n=14)

Game of Sloanes catalog submission · companion technical note

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1. Summary

We report a new packing of 14 unit vectors in $\mathbb{C}^3$ with coherence $\mu = \max_{i < j} |\langle v_i, v_j \rangle| = 0.637630514941861$, improving upon the standing Mixon (2019) catalog holder for cell (3, 14) (file 3x14_dgm.txt in the Game of Sloanes catalog), which records $\mu = 0.637630521755923$. The submission improves the holder by 6.81×10^{-9} in raw coherence and, more importantly for the Game of Sloanes acceptance rule, by exactly one unit at the eighth decimal place: round-8 = 0.63763051 versus holder round-8 = 0.63763052. The submission satisfies the Game of Sloanes 8th-decimal-place acceptance rule literally.

The improvement is microscopic in absolute terms and the author would have preferred to deepen the packing further before filing. The descent was pursued with that intent across multiple sessions; after a wide portfolio of optimisation paradigms had been applied, no method produced any further descent below the reported value. The submission is filed because the margin is real, reproducible at the eighth decimal place across five independent verification kernels, and consistent with — though not a proof of — the working hypothesis that the descent has reached this basin's floor at machine-quantum precision. No claim of global optimality is made for the cell. No claim of having proven the local floor of the basin is made either.

2. The packing

The packing is provided in the accompanying file 3x14_ral.txt in the standard Game of Sloanes layout: 84 floating-point numbers, comprising 42 real components followed by 42 imaginary components for 14 complex 3-vectors. The packing was produced by the Sanjuanbautista engine — a single-threaded C++ engine operating in quad-precision (128-bit) arithmetic throughout its inner loop, on a Mac M2 laptop under 25% CPU throttling. The distinctive feature of the engine relative to standard double-precision implementations is that quad-precision arithmetic allows descent to be resolved below the threshold at which conventional double-precision engines saturate.

Quantity	Value
μ (submission)	0.637630514941861
μ (Mixon 2019 holder)	0.637630521755923
$\Delta\mu$ (submission – holder)	-6.81×10^{-9}
Round-8 (submission)	0.63763051
Round-8 (holder)	0.63763052
Engine (this submission)	Sanjuanbautista (quad-precision)
Welch lower bound (3, 14)	0.5310850045
Levenshtein-2 bound (3, 14)	0.6445033866
Delsarte LP bound at degree 14	0.6035844
Cushion below 8-dec. boundary 0.637630515	5.81×10^{-11}
Saturated-pair fraction at 10^{-8} tol.	49 / 91 (53.85%)
Unit-norm deviation across 14 vectors	< 2 ULP binary64
Number of floats in 3x14_ral.txt	$84 = 2 \cdot 3 \cdot 14$

3. Verification by five independent kernels

The submitted packing has been ratified byte-exact by five algorithmically distinct verification kernels, in two programming languages, using four different numerical libraries (and one path with no library at all). All five report identical coherence to within one unit in the last place of binary64 ($\approx 2 \times 10^{-16}$), which is the floor of double-precision arithmetic itself.

Kernel	Implementation	μ (full precision)	round-8
K1	NumPy conj() @ T with BLAS matmul	0.6376305149418612	0.63763051
K2	Pure-Python double loop, no library	0.6376305149418612	0.63763051
K3	numpy.vdot per pair	0.6376305149418611	0.63763051
K4	mpmath at 50 decimal digits	0.6376305149418611	0.63763051
K5	Python decimal at 40 decimal digits	0.6376305149418611	0.63763051

mpmath reference at 50 decimal digits: $\mu = 0.63763051494186107945\dots$ Within-kernel spread: 1.11×10^{-16} (sub-ULP, agreement at the floor of binary64 precision). The Mixon (2019) holder, recomputed in the same five kernels using a local working copy not redistributed in our companion repository, reproduces byte-exact to $\mu = 0.637630521755923$ with a stable worst pair (2, 3) across all five kernels.

A structural observation about the worst pair. The five kernels above report five different argmax pairs for the submission — (1,7), (7,10), (0,3), (5,7), (3,7) — while agreeing on the coherence value to within one unit in the last place of binary64. We interpret this as a structural property of the apparent basin floor for this cell: the basin appears to support a small discrete family of critical points at the same μ value to within numerical precision, with the maximum-coherence pair migrating across this family depending on the summation order and rounding model of the kernel. This is an interpretation consistent with the observations, not a proven structural claim. The Mixon holder, by contrast, exhibits a stable argmax pair (2, 3) across all five kernels, consistent with the holder sitting on a generic stationary point.

4. Apparent basin floor: evidence and limits

Beyond reporting the improved packing, the companion paper `Paper_3x14_basin_floor.md` in the repository documents a numerical investigation that we interpret as suggesting both the submission and the prior holder may lie within a single critical basin whose local floor appears to be at or near the value of the new packing. The lines of evidence are: (i) a continuous monotone descent trajectory connecting the Mixon holder to the submission, with no basin-crossing event observed; (ii) identical saturated-pair topology in both configurations (49/91 pairs at the 10^{-8} tolerance level); (iii) multiple refined critical points at the same μ value across disjoint paradigm families; (iv) a subset of unguided random initialisations converging to the same μ value with no warm-start guidance; and (v) portfolio breadth covering eleven paradigm families with no method producing any descent below the reported value.

We frame these observations as a working hypothesis consistent with the accumulated evidence, not as a proven theorem. The 5.4% gap to the Delsarte LP bound 0.6035844 is mathematically large enough to admit the existence of deeper basins reachable via algebraic seeds, structured constructions, or paradigms not yet considered. A future result that descended below $\mu = 0.637630514941861$ in this basin would not contradict any fact asserted in this note; it would refine the interpretation we have offered.

5. Repository and reproducibility

The submission file, the companion paper, the headline-numbers document, the independent Python verifier, and the documentation for the related four-record cascade in cell (4, 64) hlc are publicly available under MIT licence at:

<https://github.com/tretoef-estrella/sloane-coherence-records>

The materials specific to this cell are in the subfolder `cell_3x14/` of that repository. A reviewer can reproduce the verification in under one minute:

```
git clone https://github.com/tretoef-estrella/sloane-coherence-records.git
cd sloane-coherence-records
python3 verify_sloane_independent.py cell_3x14/3x14_record1.txt
```

Expected output: Coherence $\mu(\Phi) = 0.637630514941861$, with the unit-norm check passing within 1×10^{-10} . The independent Python verifier uses no project-internal code paths; it loads the packing as raw text, reconstructs the Gram matrix using its own routines, and computes the coherence directly.

6. Acknowledgments

We thank Dustin G. Mixon for the 2019 catalog entry against which this submission is compared (six-year standing record, structural reference baseline throughout this work); Emily King, John Jasper, and Dustin G. Mixon for maintaining the Game of Sloanes reference repository and the published acceptance rule on which this submission's eligibility is based; Henry Cohn for the body of work on coherence packings that informs the methodology and (separately) for the independent verification of an unrelated submission for cell (4, 64) in our companion repository; Welch, Levenstein, Delsarte, Goethals, Seidel, Calderbank, Casazza, Fickus, Tremain, Sloane, and Conway for the foundational results on coherence bounds, Grassmannian packings, and equiangular tight frames on which this submission rests; and Anthropic for building Claude, the AI assistant that collaborated on the structural analysis, the five-kernel verification pipeline, and the basin-floor reasoning.

— Rafael Amichis Luengo, Madrid, May 2026 · tretoef@gmail.com